

Multiplication Intro

Session 1 · What Does Multiplication Mean?

■ Subject	Mathematics
■ Grade	Grade 2
■ Duration	45 – 60 minutes
■ Context	Indian curriculum, Indian examples

■ Learning Objectives

By the end of this session, students will be able to:

- Understand that multiplication means putting **equal groups** together
- Connect **repeated addition** to multiplication sentences
- Read and write a multiplication sentence using the $\times$ symbol
- Identify and describe **arrays** (rows & columns)
- Recognise that **order doesn't matter** (commutative property)

SECTION 1 · Introduction — The Superpower of Maths!

Warm-Up Story

Riya and Aarav are helping set up for Diwali. There are 4 plates of laddoos on the table — and their Amma asks: **How many laddoos are there in total?** They start counting one by one... 1, 2, 3... 4... 8... *this is taking forever!*

Then Aarav says: *Wait — I know a FASTER way!* That faster way is **Multiplication** — today's superpower! ■

■ Multiplication is the SUPERPOWER of Maths

- It lets us count equal groups very quickly
- It's just a faster way to add the same number again and again

SECTION 2 · Equal Groups

What Are Equal Groups?

When every group has the **same number** of things, we call them **equal groups**. Look at the mangoes below — they are arranged in 3 bowls with **4 mangoes in each bowl**.

Bowl 1	■ ■ ■ ■
Bowl 2	■ ■ ■ ■
Bowl 3	■ ■ ■ ■
Total →	12 mangoes

- Multiplication means putting equal groups together
- 3 groups of 4 means '3 times 4 mangoes'
- We write it as: $3 \times 4 = 12$

SECTION 3 · Repeated Addition

Repeated Addition → Multiplication

Before we knew multiplication, we used **repeated addition** — adding the same number over and over. Multiplication is just a shortcut for exactly that!

Repeated Addition	Multiplication	Same?
$4 + 4 + 4$	3×4	■ Yes! Both = 12
$5 + 5$	2×5	■ Yes! Both = 10
$3 + 3 + 3 + 3$	4×3	■ Yes! Both = 12

***Key Insight:** The multiplication sentence is **FASTER** to write and faster to solve once you know your tables!*

SECTION 4 · The Multiplication Sentence

Reading the 'x' Symbol — The TIMES Star

The symbol $\times$ is called the *TIMES star*. It means **groups of** or **times**. Every multiplication sentence has three parts:

Part	Symbol / Number	Meaning	Example ($3 \times 4 = 12$)
Number of groups	3	How many groups?	3 bowls
TIMES star	$\times$	'times' / 'groups of'	groups of
Number in each group	4	How many in each group?	4 mangoes each
Equals sign	=	Total	equals
Total / Product	12	The final answer	12 mangoes

How to READ a multiplication sentence:

Written form	Read aloud as
$2 \times 5 = 10$	"Two times five equals ten"
$3 \times 4 = 12$	"Three times four equals twelve"
$4 \times 3 = 12$	"Four times three equals twelve"

SECTION 5 · Indian Context Examples

Multiplication is everywhere in daily life! Let's see how our friends use it with familiar Indian items:

Friend	Situation	Multiplication
■ Priya	2 bags, 3 tomatoes each	$2 \times 3 = 6$ tomatoes
■ Rohan	4 teams, 5 students each	$4 \times 5 = 20$ students
■ Meera	$5 + 5 + 5$ (3 groups of 5)	$3 \times 5 = 15$
■ Amma	3 boxes of laddoos, 4 each	$3 \times 4 = 12$ laddoos
■ Nani	2 idli tiffins, 5 idlis each	$2 \times 5 = 10$ idlis

SECTION 6 · Arrays — Rows and Columns

Seeing Multiplication as an Array

An **array** is when objects are arranged in neat **rows** (going across ➡) and **columns** (going down ▼). Arrays make it easy to see multiplication.

	Col 1	Col 2	Col 3	Col 4	Col 5
	■	■	■	■	■
Row 2	■	■	■	■	■
	■	■	■	■	■
Row 4	■	■	■	■	■

4 rows × 5 columns = 4 × 5 = 20 eggs ■

The Commutative Property — Order Doesn't Matter!

Here's something magical: you can **swap the numbers** in a multiplication sentence and the answer stays the same. This is called the **Commutative Property**.

4 × 5 = 20	↔	5 × 4 = 20
(4 rows of 5)		(5 rows of 4)
Same total, different arrangement! ■		

SECTION 7 · Practice Questions

Let's test what we've learned! Try to answer each question on your own before checking the answers.

Q1. What does 3 × 2 mean?

- a) 3 + 2
- b) 3 groups of 2
- c) 3 – 2

Answer: (b) 3 groups of 2 — The × symbol means 'groups of', so 3 × 2 = three groups of two.

Q2. Fill in the blanks: 2 + 2 + 2 + 2 = ____ × ____

Answer: 4 × 2 — The number 2 is added 4 times, so it is 4 groups of 2.

Q3. Aarav has 3 jars of candies with 5 candies in each jar. Write the multiplication sentence.

Answer: $3 \times 5 = 15$ — 3 groups, 5 in each group $\rightarrow 3 \times 5 = 15$ candies.

Q4. An egg tray has 4 rows and 5 columns. How many eggs in total?

Answer: $4 \times 5 = 20$ — Rows $\times$ columns = $4 \times 5 = 20$ eggs.

Q5. Riya says $3 \times 6 = 18$. Meera says $6 \times 3 = 18$. Who is correct?

- a) Riya only
- b) Meera only
- c) Both are correct

Answer: (c) Both are correct — The commutative property says: $a \times b = b \times a$. Both 3×6 and 6×3 equal 18.

Q6. (Challenge) Draw an array for 2×4 . Then write the matching repeated addition sentence.

Answer: $4 + 4 = 2 \times 4 = 8$ — Array: 2 rows of 4 objects. Repeated addition: $4 + 4 = 8$.

SECTION 8 · Recap & Summary

■ What We Learned Today

- ■ Multiplication = equal groups put together
- ■ It is the FASTER way to add the same number repeatedly
- ■ Write as: [groups] $\times$ [in each group] = [total]
- ■ The $\times$ symbol is called the TIMES star
- ■ Arrays show multiplication as rows and columns
- ■ Commutative property: $a \times b = b \times a$ (order doesn't change the answer)

SECTION 9 · Fun Challenge at Home! ■

Look around your home and spot **equal groups** everywhere! When you find them, write out the multiplication sentence and tell your family.

Item	Example situation	Your multiplication sentence
■ Biscuits in a packet	4 rows of 5 biscuits	$4 \times 5 = \underline{\quad}$
■ Windows in a building	3 floors, 4 windows each	$3 \times 4 = \underline{\quad}$
■ Bangles in a box	2 rows of 6 bangles	$2 \times 6 = \underline{\quad}$
■ Eggs in a tray	3 rows of 4 eggs	$3 \times 4 = \underline{\quad}$
■ Books on a shelf	2 shelves, 7 books each	$2 \times 7 = \underline{\quad}$

Tip for parents: Encourage your child to say the sentence aloud, e.g. "Two times six equals twelve!"
Verbalising builds confidence.

■ Multiplication Star — Session 1

Complete! ■

Great job! You've completed Session 1. See you in Session 2 — where we explore the
Multiplication Tables! ■

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